mistake, on average. We should be rooting for machines that make fewer errors, not excusing human error.

Even if one buys the argument that models can be useful, one might object that models are too limited and cannot be applied in sophisticated contexts like investment decision-making. What, for example, is the algorithm going to say when we face a unique situation the world has never seen? This time is *actually* different. The story is that the human expert can adapt and create on-the-fly modifications to the model that creates value. This well-trodden, but empirically busted, rebuttal against algorithms is deemed the *broken-leg theory* and relies on the false premise that humans don't suffer from System 1 flaws.

Under the broken leg theory, if we are competing against a model to judge whether someone will, say, go to the movies, and we happen to know this person has a broken leg and the model does not account for this, then we can use this unique, incremental knowledge to make a better judgment than the machine. One problem with this view is that it is difficult to know whether a given information signal (which may be more nuanced than a broken leg) is dispositive and reliable. Thus, we continue to extend and apply our judgment over and over again in different cases, and override the model when we shouldn't.

Sure, knowledge of the broken leg helps us beat the model. Discretionary decision makers are often able to identify the value-enhancing modifications that can theoretically outperform a simple model in specific cases. However, they simultaneously identify value-destroying modifications that cause them to underperform in other cases. Discretionary experts' inclination to “modify” simple models resembles a bag of Lay's Potato Chips—the experts “can't eat just one” modification. They might add a positive “broken leg” modification, but they also add negative modifications. And as the evidence suggests, allowing a human to engage in ad-hoc “gut” decision-making is not a good idea.

Of course, the great irony is that an expert can read and understand the evidence on systematic versus ad-hoc decision-making and agree that decisions should be made systematically with a simple model. However, overconfidence, arguably the bias that hurts experts the most, causes these